

Global Tuberculosis Burden & Treatment Gaps

Likely questions, defensible answers and wording guardrails

Q&A PREPARATION

A decision-support tool for understanding where tuberculosis burden is concentrated, how trends have changed, and where diagnosis-and-treatment coverage remains incomplete.

Name	[Student Name]
Major	Master of Science in Business Analytics
Department	Suliman S. Olayan School of Business
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Project snapshot Analytical period 2000–2024 · 217 countries/territories · six WHO regions · age/sex cross-section for 2024 · reproducible source manifest and validation checks.

Prepared as an individual master's-level healthcare analytics project

Q&A STRATEGY

Answer in three steps

Step	What to do	Example
1. Answer	Give the direct answer in one sentence.	"No, the dashboard does not prove causality."
2. Evidence	Support it with one method or number.	"It uses ecological country-level data."
3. Boundary	State the limitation or next step.	"A causal study would need individual or quasi-experimental data."

Wording guardrails

- Say "estimated incident cases," not "confirmed cases."
- Say "notified new and relapse cases," not simply "treated patients."
- Say "estimated notification difference" or "coverage signal," not "all untreated people."
- Say "exploratory projection," not "prediction of what will happen."
- Say "associated geographic pattern," not "cause."

Best fallback "The dashboard is designed to prioritize investigation. It does not replace programme validation or clinical judgment."

LIKELY QUESTIONS · PART 1

Concept and interpretation

What exactly is the “treatment gap”?

The dashboard's core measurable gap is diagnosis-and-treatment coverage: notified new and relapse cases divided by estimated incident cases. The absolute difference is called the estimated notification difference. I avoid claiming that every person in the difference was untreated because incidence is estimated and notifications are programme counts.

Why is TB related to HIV?

HIV weakens immunity, increasing the chance that TB infection progresses to active disease. They are different diseases, so TB–HIV is analysed as a co-epidemic requiring integrated testing and care.

What is RR/MDR-TB?

RR-TB is resistant to rifampicin, a key TB medicine. MDR-TB is resistant to at least rifampicin and isoniazid. These cases require rapid resistance testing and different treatment pathways.

Why use both counts and rates?

Counts identify where the largest service volume is needed. Rates standardize for population size and identify intense risk. Coverage adds the health-system reach dimension.

Data and methodology

Where did the data come from?

The project uses WHO Global Tuberculosis Programme indicators. The package includes raw WHO-derived snapshots, an official WHO report/repository cross-check, source URLs, retrieval dates and SHA-256 hashes.

Why did you use a mirror rather than the direct WHO CSV?

The WHO data page is the authoritative source, but its direct CSV endpoint was not retrievable in the development runtime. I used a public WHO-derived DDF mirror, retained full provenance, and validated results against the current WHO report and official 2025 repository.

Why are your global totals 10.55M and 1.20M, while WHO says 10.7M and 1.23M?

WHO publishes rounded global estimates with uncertainty intervals and updates country time series. My figures are reproducible sums of the country-level snapshot. The dashboard discloses this difference instead of presenting the two as identical.

What data cleaning did you perform?

I standardized country identifiers and names, converted years and indicators to numeric form, merged 15 indicator files, recalculated rates and coverage, selected non-overlapping age bands, and created country, regional and global tables.

How did you validate the data?

Automated checks tested unique country-year records, year range, non-negative burden values, ISO-code completeness, expected WHO regions, and the presence of male and female age-sex estimates.

LIKELY QUESTIONS · PART 2

Analysis, prediction and limitations

Why can calculated coverage exceed 100%?

Incidence is estimated and notifications are reported. Different uncertainty, timing, revisions or cross-border diagnosis can produce a numerator above the estimated denominator. The dashboard preserves the value but uses a non-negative gap field for ranking.

Why use a simple linear forecast?

Prediction is optional bonus work. I chose an interpretable baseline, report holdout MAE, compare it with a last-value baseline, and label the result as an exploratory extrapolation rather than a policy or clinical forecast.

Why is age/sex only for 2024?

That is the available cross-sectional age-sex estimate in the integrated project package. I state the limitation and do not imply a demographic time trend.

Why not include treatment success outcomes?

The current version focuses on burden, notifications and diagnosis-and-treatment coverage. Treatment cohorts use different reference years and require careful alignment. They are a logical extension, but omitting them keeps this version methodologically coherent.

Can the dashboard prove causes?

No. It uses ecological country-level data and supports prioritization, not causal inference or individual risk prediction.

What is the single most important finding?

Progress is visible, but the response is uneven: a global 78.9% calculated coverage still leaves a large notification difference, and priority changes depending on burden volume, rate, HIV co-burden and drug resistance.

What would you improve next?

I would add treatment-cohort outcomes, uncertainty intervals, subnational data for selected countries, and stronger forecast models only after verifying sufficient and comparable time-series data.

Source references to mention

- WHO Global tuberculosis report 2025 and its top findings.
- WHO Tuberculosis data page and data dictionary.
- GTB-TME official 2025 report GitHub repository.
- Project data manifest, preparation script and validation JSON.